

Multiple Choice (10 marks)

1. **The theory of evolution is most accurately described as**
 - (a) a universally accepted fact about the origin and development of all species.
 - (b) a hypothesis in the process of being tested and verified.
 - (c) a disproven theory about the development of species over time.
 - (d) a speculation about possible changes in populations over time.
 - (e) an opinion that some scientists hold about how living things change over time.
2. **Why is genetic drift less important in human evolution at present than it was twenty thousand or more years ago?**
 - (a) The human genome has fully adapted to current environmental pressures
 - (b) Modern medicine has eliminated the effects of genetic drift
 - (c) The rate of genetic drift is directly proportional to technological advancements
 - (d) Humans have reached a genetic equilibrium where drift is no longer possible
 - (e) Humans have developed immunity to genetic drift
3. **A male bacterium conjugates with a female bacterium. After conjugation, the female becomes a male. Account for this "sexchange".**
 - (a) The female bacterium becomes a male as a result of nutrient deficiency
 - (b) The female becomes a male by receiving one copy of the F factor
 - (c) The female bacterium becomes a male through the process of osmosis
 - (d) The female bacterium becomes a male through gene mutation
 - (e) The female bacterium becomes a male after receiving multiple copies of non-F factor genes
4. **To prevent desiccation and injury, the embryos of terrestrial vertebrates are encased within a fluid secreted by the**
 - (a) blastoderm
 - (b) mesoderm
 - (c) amnion
 - (d) zona pellucida
 - (e) ectoderm
5. **Why are the stem reptiles especially important in evolution?**
 - (a) They exclusively evolved into modern reptiles without any other descendants

- (b) They were the first reptiles to return to an aquatic lifestyle
- (c) They gave rise to only the dinosaurs
- (d) They were the only reptiles to develop flight
- (e) They existed only in the Mesozoic Era

True / False (10 marks)

Answer True or False

- 6. True or False: Sister chromatids separate during meiosis II, not meiosis I, which involves the separation of homologous chromosomes.
- 7. True or False: The acrosome and cortical reactions allow multiple sperm to fertilize the egg, leading to polyembryony.
- 8. True or False: The drop in blood pressure can lead to blood clotting due to reduced shear stress.
- 9. True or False: An extra Y chromosome (XXY) is linked to an increase in cognitive abilities and intelligence.
- 10. True or False: Xylem vessel members undergo programmed cell death to facilitate water transport.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms by which the pharynx directs food to the esophagus and air to the trachea during swallowing, emphasizing the roles of the epiglottis and neural reflexes.
- 12. Discuss the changes that occur in the A band of a sarcomere during muscle contraction, and analyze how these changes contribute to the overall process of muscle shortening.

End of Paper